Huangming Wang

LLM Application Engineer

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Summary

LLM application engineer with 2+ years of hands-on experience building AI-enabled and full-stack software through internships and project work. Develops reliable, domain-specific systems across RAG, agentic workflows, local and cloud LLM integration, structured tool use, evaluation, and end-to-end application delivery. Applies AI responsibly through grounded generation, schema validation, source provenance, traceable execution, privacy-conscious local inference, and deterministic controls around model actions.

Skills

AI and LLM Engineering: RAG, context and prompt engineering, ReAct agents, multi-agent workflows, tool calling, structured outputs, evaluation, SFT, LoRA/QLoRA, LangChain, LangGraph, Ollama, OpenAI, Gemini, Groq

Knowledge and Data Systems: Markdown-first knowledge systems, Qdrant, Chroma, Neo4j, MongoDB, SQLite, SQL, embeddings, semantic search, PDF ingestion, source provenance, citation and schema validation

ML and Analytics: Regression, classification, clustering, time series analysis, random forest, pandas, PyTorch, evaluation harnesses, backtesting

Application Engineering: Python, JavaScript/TypeScript, Java, Kotlin, C++, FastAPI, Express.js, Spring Boot, React, Redux Toolkit, PySide6, Android, REST APIs, Docker, Git, CI/testing

Work Experience

AI Engineer Intern, Coinhome 2026 - Present

- Developed CryptoGPT and MATS Harness, refactoring a fixed multi-agent trading workflow into per-team ReAct loops with typed tools, modular strategy skills, and full-cycle JSONL traces.
- Separated backtest, paper, and controlled live execution; integrated explicit live confirmation, halt controls, capital validation, MongoDB order records, CCXT market access, and an operational dashboard.
- Architected a local Markdown-first RAG Wiki that compiles source documents into maintainable, cited knowledge pages through staged extraction, page planning, and drafting workflows.
- Built Plan/Preview/Apply safeguards with source hashing, citation validation, stale-plan detection, transactional writes, rollback, deterministic evaluation, and an optional Neo4j graph projection.

Software Engineer Intern, Neutron Mobile May 2025 - Sep 2025

- Developed LawGPT, a legal AI chatbot using RAG to ground answers in uploaded legal documents and support context-aware conversations.
- Built Express.js and MongoDB APIs for authentication, chat history, document processing, and model selection; integrated PDF ingestion, embeddings, and Qdrant retrieval.
- Supported OpenAI, Gemini, Groq, and Ollama providers and delivered a React/Vite/Redux interface with protected routes, persistent conversations, and real-time chat UX.

Projects

MATS Harness - Multi-Agent Trading System

Python, ReAct, LangGraph, MongoDB, CCXT, Docker, React dashboard

- Designed a production-style harness for crypto spot trading agents with per-team ReAct loops, tool registry, skill loading, and full-cycle trace logging under `runs/[run_id]/trace.jsonl`.
- Created paper and live execution boundaries with explicit live confirmation, kill switch controls, risk checks, and order ledger persistence to keep research and execution separated.
- Built independently testable strategy Skills and CLI/dashboard workflows for backtesting, live traces, chat/tool replay, portfolio state, order inspection, and Docker-based deployment.

LawGPT - Legal RAG Chatbot Platform
React, Vite, Redux Toolkit, Express.js, MongoDB, Qdrant, LangChain, Ollama

- Implemented full-stack legal assistant with user authentication, conversation persistence, document ingestion, vector search, and context-aware chat responses.
- Built document embedding pipeline for PDFs: parse, chunk, embed, store in Qdrant, retrieve semantically relevant passages, and inject context into model prompts.
- Added cloud/local model routing and application controls including JWT auth, email verification, password reset, Google OAuth/reCAPTCHA, secure cookies, CORS, and admin API protection.

GridScribe - Power-Grid Speech Correction API
FastAPI, Whisper, Ollama, local RAG, SQLite, Pydantic, QLoRA training pipeline

- Designed a privacy-preserving inspection pipeline: audio validation, ffmpeg preprocessing, local Whisper transcription, local RAG retrieval, Ollama correction, JSON extraction, and schema validation.
- Built multi-format RAG ingestion and prepared human-reviewed examples for future SFT/QLoRA without treating unreviewed model outputs as training labels.

Game Companion - Local Game Knowledge Agent (In Development)
Python, PySide6, Ollama, Gemma 4B, Pydantic, Markdown, PyInstaller

- Developing a private, spoiler-safe desktop Agent that organizes player notes into a searchable and editable Markdown workspace using local edge-model inference.
- Replaced large one-shot outputs with a recoverable tool loop using schema validation, source-coverage checks, atomic writes, archives, and batch rollback; established 273 passing offline tests.

AI Recipe Generation and Ingredient Management
Kotlin, Android, Spring Boot, Docker, Llama 3.1, Chroma

- Built an Android and Spring Boot ingredient-management application with Dockerized Llama 3.1 generation and Chroma-backed retrieval across Kotlin, Java, and Python services.

Certifications

Cisco Certified Network Associate (CCNA), Cisco — May 2024 - Aug 2024
Routing, switching, IP services, network security, and automation fundamentals.

Education

DePauw University, Greencastle, IN — Aug 2022 - May 2026
Bachelor of Arts in Computer Science, Minor in Mathematics
GPA 3.85/4.00

- Dean's List: Fall 2022 - Fall 2025
- Computer Science Honor Society